

COMBINATORICS

EXAM 1

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1. Answer the following using simple counting reasoning

- (a) How many different 7-digit numbers (non-negative integers) are there? (Leading zeros are not allowed).
- (b) How many even 7-digit numbers are there?
- (c) How many 7-digit numbers are there with exactly one 7?
- (d) How many 7-digit numbers read the same when you read them forward or backward, i.e. are palindromic, e.g. 6548456 is one such integer.
- (e) How many 7-digit numbers are there that are divisible by 4?
- (f) How many times is the digit 7 written when listing the numbers between 1 and 1,000,000?

Solution.

□

2. How many 7-letter words can be constructed by using the 26 letters of the alphabet if each word contains 3, or 4 vowels? It is understood that there is no restriction on the number of times a letter can be used in a word.

Solution.

□

3. Let n and k be positive integers. Give a combinatorial proof that

$$\sum_{k=1}^n k \binom{n}{k}^2 = n \binom{2n-1}{n-1}$$

Proof.

□

4. In how many ways can we select a set of m marbles from a collection of $2m$ marbles, where m of them are identical and the remaining m are all distinct?

Solution.

□

5. Show that if $n + 1$ distinct integers are chosen from the set $\{1, 2, \dots, 3n\}$, then there are always two which differ by at most 2.

Proof.

□

6. Use the pigeonhole principle to show that:

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- (a) Any set of 7 distinct integers includes two integers a and b such that either $a + b$ or $a - b$ is a multiple of 10.
- (b) Any group of 20 students either has a subgroup of 4 mutual acquaintances, or a subgroup of 4 mutual strangers.

Solution.

□

7. What is the number of integral solutions of the equation

$$x_1 + x_2 + x_3 + x_4 = 30,$$

such that $x_1 \leq 1$, $x_2 \leq 2$, $x_3 \leq 3$, and $x_4 \leq 4$?

Solution.

□

8. Find the number of integer solutions to the equation

$$x_1 + x_2 + x_3 + x_4 = 20,$$

for which $1 \leq x_1 \leq 6$, $0 \leq x_2 \leq 7$, $4 \leq x_3 \leq 8$, and $2 \leq x_4 \leq 6$?

Solution.

□

9. Solve the recurrence relation

$$a_n = 5a_{n-1} + 2a_{n-2} - 24a_{n-3}, \quad n \geq 3$$

given that $a_0 = 1$, $a_1 = 2$, and $a_2 = 6$.

Solution.

□

10. Solve the recurrence relation $a_k = a_{k-1} + k + 6$, with $k \geq 1$ given that $a_0 = 5$.

Solution.

□

11. Use generating functions to count the number of ways to collect \$15 from 20 different students, if each of the first 19 students can give either \$1 or nothing, and the twentieth student can give nothing, \$1, or \$5.

Solution.

□

12. Consider an n -sided polygon (with n vertices and n sides).

- (a) Assuming no three diagonals intersect at a single point, count the points of intersection formed by the diagonals of this n -sided polygon.
- (b) How many line segments are formed by the intersection of points in part (a)?

- (c) One of the simplest and most elegant formulae in mathematics is Euler's polyhedron formula. A version for planar graphs states that if a planar graph with v vertices and e edges is drawn in the plane without any edge crossings and has r regions, then

$$r = e - v + 2$$

Use this formula together with your answers in parts (a) and (b) to count the number of regions formed by the diagonals of the n -sided polygon.

Solution.

□

- 13.** Show that any set of 3 distinct integers includes two integers x and y such that

$$F(x, y) = x^3y - xy^3$$

is a multiple of 10.

Proof.

□

- 14.** Use combinatorial reasoning to show that:

$$\binom{n}{r-1} = \sum_{i=0}^k (-1)^i \binom{k}{i} \binom{n+k+1}{r+k+1}$$

Proof.

□

- 15.** Find and verify a general formula, involving the Sterling Numbers of the second kind, for $1^n + 2^n + 3^n + \cdots + m^n$.

Solution.

□